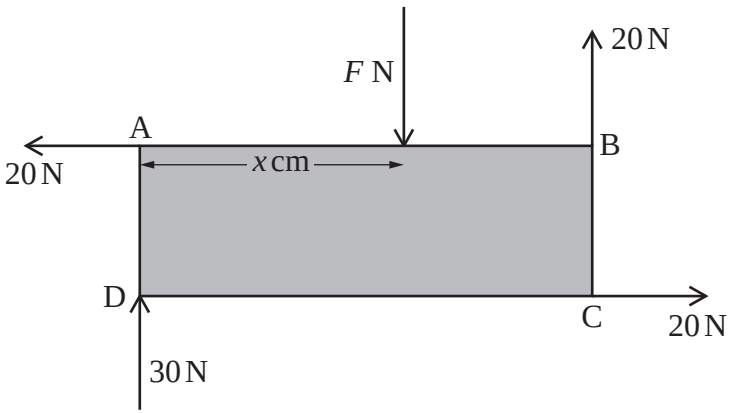
- The acceleration due to gravity is denoted by $g\,\text{m s}^{-2}$. When a numerical value is needed use $g = 9.8$ unless a different value is specified in the question.

5 ABCD is a rectangular lamina in which AB is 30 cm and AD is 10 cm.

Three forces of 20 N and one force of 30 N act along the sides of the lamina as shown in the diagram.



An additional force F N is also applied at right angles to AB to a point on the edge AB x cm from A.

(a) Given that the lamina is in equilibrium, calculate the values of F and x. **[3]**

The point of application of the force F N is now moved to B, but the magnitude and direction of the force remain the same.

(b) Explain the effect of the new system of forces on the lamina. **[2]**

5(a)	
	$F =$
	$x =$
5(b)	